

Операционные системы

Основы интерфейса взаимодействия пользователя с системой Unix на уровне командной строки

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Приобретение практических навыков взаимодействия пользователя с системой посредством командной строки.

- 1 Определить имя и путь домашнего каталога
- 2 Изучить команду `ls`.
- 3 Выполнить действия с каталогами.
- 4 Получить дополнительные сведения при помощи справки по командам.
- 5 Изучить команду `history`.

2. Процесс выполнения лабораторной работы

```
samalutina@samalutina:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$  
samalutina@samalutina:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ cd  
samalutina@samalutina:~$ pwd  
/home/samalutina  
samalutina@samalutina:~$
```

Рисунок 1: Путь к домашнему каталогу

```
samalutina@samalutina:~$ cd /tmp
samalutina@samalutina:/tmp$ ls
11bc2e78-67ed-4c1e-a0c3-076bdc1e2d4.zip
8781fa62-18d7-40fa-bde1-419130519051.zip
happd.sock
hsperfdata_samalutina
node-compile-cache
par-73616d616c7574696e61
snap-private-tmp
systemd-private-2a207e477e734273afdd53be67b3c110-abrt.service-53y1Q0
systemd-private-2a207e477e734273afdd53be67b3c110-chrond.service-lv1db0
systemd-private-2a207e477e734273afdd53be67b3c110-colord.service-UkFUMc
systemd-private-2a207e477e734273afdd53be67b3c110-dbus-broker.service-CuT7gQ
systemd-private-2a207e477e734273afdd53be67b3c110-irqbalance.service-sq6xZk
systemd-private-2a207e477e734273afdd53be67b3c110-low-memory-monitor.service-8q0XeW
systemd-private-2a207e477e734273afdd53be67b3c110-ModemManager.service-9UV8i5
systemd-private-2a207e477e734273afdd53be67b3c110-pcscd.service-s6lba1
systemd-private-2a207e477e734273afdd53be67b3c110-polkit.service-LBsZnX
systemd-private-2a207e477e734273afdd53be67b3c110-rtkit-daemon.service-NhBQvq
systemd-private-2a207e477e734273afdd53be67b3c110-switcheroo-control.service-ycThvv
systemd-private-2a207e477e734273afdd53be67b3c110-systemd-logind.service-l9w4VE
systemd-private-2a207e477e734273afdd53be67b3c110-systemd-resolved.service-2bSYia
systemd-private-2a207e477e734273afdd53be67b3c110-upower.service-rHSuby
VMwareDnD
vmware-root_896-2730693439
samalutina@samalutina:/tmp$
```

Рисунок 2: Команда ls

Опции команды ls

```
samalutina@samalutina:/tmp$ ls -la
.
..
11bc2e78-67ed-4c1e-a0c3-076bdcbb1e2d4.zip
8781fa62-18d7-40fa-bde1-419130519051.zip
.font-unix
happd.sock
hsperfdata_samalutina
.ICE-unix
node-compile-cache
par-73616d616c7574696e61
snap-private-tmp
systemd-private-2a207e477e734273afdd53be67b3c110-abrtd.service-53y1Q0
systemd-private-2a207e477e734273afdd53be67b3c110-chrond.service-iv1db0
systemd-private-2a207e477e734273afdd53be67b3c110-colord.service-UkFUMc
systemd-private-2a207e477e734273afdd53be67b3c110-dbus-broker.service-CuT7gQ
systemd-private-2a207e477e734273afdd53be67b3c110-irqbalance.service-sq6xZk
systemd-private-2a207e477e734273afdd53be67b3c110-low-memory-monitor.service-8q0XeW
systemd-private-2a207e477e734273afdd53be67b3c110-ModemManager.service-9UV8i5
systemd-private-2a207e477e734273afdd53be67b3c110-pcsd.service-s6lba1
systemd-private-2a207e477e734273afdd53be67b3c110-polkit.service-LBsZnX
systemd-private-2a207e477e734273afdd53be67b3c110-rtkit-daemon.service-NhBQvq
systemd-private-2a207e477e734273afdd53be67b3c110-switcheroo-control.service-ycThvv
systemd-private-2a207e477e734273afdd53be67b3c110-systemd-logind.service-l9w4VE
systemd-private-2a207e477e734273afdd53be67b3c110-systemd-resolved.service-2bSYIa
systemd-private-2a207e477e734273afdd53be67b3c110-upower.service-rHSuby
VMwareDnD
vmware-root_896-2730693439
.X0-lock
.X11-unix
.X1-lock
.XIM-unix
samalutina@samalutina:/tmp$
```

Рисунок 3: Команда ls -la

Опции команды ls

```
samalutina@samalutina:/tmp$ ls -l
total 7712
-rw-r--r--. 1 samalutina samalutina 3947343 Sep  1 10:52 11bc2e78-67ed-4c1e-a8c3-076bdcblе2d4.zip
-rw-r--r--. 1 samalutina samalutina 3947343 Sep  1 10:50 8781fa62-18d7-40fa-bde1-419130519051.zip
srw-rw-rw-. 1 root      root      0 Sep  1 10:48 happd.sock
drwxr-xr-x. 2 samalutina samalutina 40 Sep  1 11:10 hsperrdata_samalutina
drwxr-xr-x. 3 samalutina samalutina 60 Sep  1 11:18 node-compile-cache
drwx-----. 3 samalutina samalutina 60 Sep  1 11:06 par-73616d616c75746986e61
drwx-----. 2 root      root      40 Sep  1 10:48 snap-private-tmp
drwx-----. 3 root      root      60 Sep  1 10:48 systemd-private-2a207e477e734273afdd53be67b3c110-abrtd
.service-53y100
drwx-----. 3 root      root      60 Sep  1 10:48 systemd-private-2a207e477e734273afdd53be67b3c110-chron
yd.service-lvldb0
drwx-----. 3 root      root      60 Sep  1 10:48 systemd-private-2a207e477e734273afdd53be67b3c110-color
d.service-UkFUMc
drwx-----. 3 root      root      60 Sep  1 10:48 systemd-private-2a207e477e734273afdd53be67b3c110-dbus-
broker.service-CuT7gQ
drwx-----. 3 root      root      60 Sep  1 10:48 systemd-private-2a207e477e734273afdd53be67b3c110-irqba
lance.service-sq6xZk
drwx-----. 3 root      root      60 Sep  1 10:48 systemd-private-2a207e477e734273afdd53be67b3c110-low-m
emory-monitor.service-8q0XeW
drwx-----. 3 root      root      60 Sep  1 10:48 systemd-private-2a207e477e734273afdd53be67b3c110-Modem
Manager.service-9UV8i5
drwx-----. 3 root      root      60 Sep  1 10:48 systemd-private-2a207e477e734273afdd53be67b3c110-pcscd
.service-s6lba1
drwx-----. 3 root      root      60 Sep  1 10:48 systemd-private-2a207e477e734273afdd53be67b3c110-polki
t.service-LBsZnX
drwx-----. 3 root      root      60 Sep  1 10:48 systemd-private-2a207e477e734273afdd53be67b3c110-rtkit
-daemon.service-NhBQvq
drwx-----. 3 root      root      60 Sep  1 10:48 systemd-private-2a207e477e734273afdd53be67b3c110-switc
heroo-control.service-ycThvv
drwx-----. 3 root      root      60 Sep  1 10:48 systemd-private-2a207e477e734273afdd53be67b3c110-syste
ad-logged.service-l9w4VF
```

Рисунок 4: Команда ls -l

```
samalutina@samalutina:/tmp$ ls -f
.
..
node-compile-cache
hsperfdata_samalutina
par-73616d616c7574696e61
VMwareDnD
11bc2e78-67ed-4c1e-a0c3-076bdcblе2d4.zip
8781fa62-18d7-40fa-bde1-419130519051.zip
.X1-lock
.X0-lock
systemd-private-2a207e477e734273afdd53be67b3c110-pcsd.service-sG1ba1
systemd-private-2a207e477e734273afdd53be67b3c110-colord.service-UkFUMc
happd.sock
systemd-private-2a207e477e734273afdd53be67b3c110-ModemManager.service-9UV8i5
vmware-root_896-2730693439
systemd-private-2a207e477e734273afdd53be67b3c110-systemd-logind.service-l9w4VE
systemd-private-2a207e477e734273afdd53be67b3c110-upower.service-rHSuby
systemd-private-2a207e477e734273afdd53be67b3c110-switcheroo-control.service-ycThvv
systemd-private-2a207e477e734273afdd53be67b3c110-rtkit-daemon.service-NhBQvq
systemd-private-2a207e477e734273afdd53be67b3c110-polkit.service-LBsZnX
systemd-private-2a207e477e734273afdd53be67b3c110-low-memory-monitor.service-8qOXeW
systemd-private-2a207e477e734273afdd53be67b3c110-chrond.service-Iv1db0
systemd-private-2a207e477e734273afdd53be67b3c110-irqbalance.service-sq6xZk
systemd-private-2a207e477e734273afdd53be67b3c110-abrt.service-53y1Q0
systemd-private-2a207e477e734273afdd53be67b3c110-dbus-broker.service-CuT7gQ
systemd-private-2a207e477e734273afdd53be67b3c110-systemd-resolved.service-2bSYia
.font-unix
.XIM-unix
.ICE-unix
.X11-unix
snap-private-tmp
samalutina@samalutina:/tmp$
```

Рисунок 5: Команда ls -f

```
samalutina@samalutina:/tmp$ cd /var/spool/
samalutina@samalutina:/var/spool$ ls -l
total 0
drwxr-x--x. 1 root abrt 24476 Aug 31 13:37 abrt
drwx-----. 1 abrt abrt    0 Dec  4  2025 abrt-upload
drwxr-xr-x. 1 root root   66 Oct 23  2025 anacron
drwx-----. 1 root root   18 Oct 23  2025 at
drwx-----. 1 root root    0 Aug  6  2025 cron
drwx--x---. 1 root lp     6 Dec 12  2025 cups
drwxr-xr-x. 1 root root    0 Jul 30  2025 lpd
drwxrwxr-x. 1 root mail  174 Sep  1 10:48 mail
drwxr-xr-x. 1 root root    0 Jan 27  2026 plymouth
samalutina@samalutina:/var/spool$
```

Рисунок 6: Каталог /var/spool

```
samalutina@samalutina:/var/spool$ cd
samalutina@samalutina:~$ ls
Desktop  Documents  Downloads  git-extended  Music  Pictures  Public  Templates  Videos  work
samalutina@samalutina:~$ ls -al
total 24
drwx----- 1 samalutina samalutina 370 Sep  1 11:31 .
drwxr-xr-x 1 root      root      174 Sep  1 10:48 ..
-rw----- 1 samalutina samalutina 2703 Sep  1 11:33 .bash_history
-rw-r--r-- 1 samalutina samalutina  18 Jul 23 2025 .bash_logout
-rw-r--r-- 1 samalutina samalutina 144 Jul 23 2025 .bash_profile
-rw-r--r-- 1 samalutina samalutina 680 Sep  1 11:18 .bashrc
drwx----- 1 samalutina samalutina 484 Sep  1 11:24 .cache
drwx----- 1 samalutina samalutina 360 Sep  1 11:24 .config
drwxr-xr-x 1 samalutina samalutina   0 Sep  1 10:48 Desktop
drwxr-xr-x 1 samalutina samalutina   0 Sep  1 10:48 Documents
drwxr-xr-x 1 samalutina samalutina   0 Sep  1 10:48 Downloads
-rw-r--r-- 1 samalutina samalutina 334 Nov 14 2025 .emacs
-rw-r--r-- 1 samalutina samalutina 235 Sep  1 11:31 .gitconfig
drwxr-xr-x 1 samalutina samalutina  74 Sep  1 11:27 git-extended
drwx----- 1 samalutina samalutina 136 Sep  1 10:58 .gnupg
drwx----- 1 samalutina samalutina  20 Sep  1 10:48 .local
drwxr-xr-x 1 samalutina samalutina  34 Oct 23 2025 .mozilla
drwxr-xr-x 1 samalutina samalutina   0 Sep  1 10:48 Music
drwxr-xr-x 1 samalutina samalutina   0 Sep  1 10:48 Pictures
drwxr-xr-x 1 samalutina samalutina   0 Sep  1 10:48 Public
drwx----- 1 samalutina samalutina 132 Sep  1 11:01 .ssh
drwxr-xr-x 1 samalutina samalutina   0 Sep  1 10:48 Templates
drwxr-xr-x 1 samalutina samalutina  18 Sep  1 11:05 .texlive2025
drwxr-xr-x 1 samalutina samalutina   0 Sep  1 10:48 Videos
drwxr-xr-x 1 samalutina samalutina  10 Sep  1 11:00 work
samalutina@samalutina:~$
```

Рисунок 7: Файлы в домашнем каталоге

```
samalutina@samalutina:~$ mkdir newdir
samalutina@samalutina:~$ mkdir newdir/morefun
samalutina@samalutina:~$ mkdir letters memos misk
samalutina@samalutina:~$ ls
Desktop  Downloads  letters  misk  newdir  Public  Videos
Documents  git-extended  memos  Music  Pictures  Templates  work
samalutina@samalutina:~$ rm letters/ memos/ misk/
rm: cannot remove 'letters/': Is a directory
rm: cannot remove 'memos/': Is a directory
rm: cannot remove 'misk/': Is a directory
samalutina@samalutina:~$ rm -r letters/ memos/ misk/
samalutina@samalutina:~$ rm -r newdir/
samalutina@samalutina:~$ ls
Desktop  Documents  Downloads  git-extended  Music  Pictures  Public  Templates  Videos  work
samalutina@samalutina:~$
```

Рисунок 8: Действия с каталогами

```
'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_extensions/yamadha  
rma/minted-quarto':  
_extension.yml  minted-quarto.lua  
  
'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/image':  
solvay.jpg  
  
'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources':  
csl  tex  
  
'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources/csl':  
gost-r-7-0-5-2008-numeric.csl  
  
'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources/tex':  
preamble.tex  
  
'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/scripts':  
image-report  mpv-shot  
samalutina@samalutina:~$ ls -t  
git-extended  work  Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  
samalutina@samalutina:~$
```

Рисунок 9: Команда ls -R и ls -t

```
samalutina@samalutina:~$ help cd
cd: cd [-L|[-P [-e]]] [-@] [dir]
    Change the shell working directory.

    Change the current directory to DIR.  The default DIR is the value of the
    HOME shell variable.  If DIR is "-", it is converted to $OLDPWD.

    The variable CDPATH defines the search path for the directory containing
    DIR.  Alternative directory names in CDPATH are separated by a colon (:).
    A null directory name is the same as the current directory.  If DIR begins
    with a slash (/), then CDPATH is not used.

    If the directory is not found, and the shell option `cdable_vars' is set,
    the word is assumed to be a variable name.  If that variable has a value,
    its value is used for DIR.

Options:
  -L      force symbolic links to be followed: resolve symbolic
          links in DIR after processing instances of `..'
  -P      use the physical directory structure without following
          symbolic links: resolve symbolic links in DIR before
          processing instances of `..'
  -e      if the -P option is supplied, and the current working
          directory cannot be determined successfully, exit with
          a non-zero status
  -@      on systems that support it, present a file with extended
          attributes as a directory containing the file attributes

    The default is to follow symbolic links, as if `-L' were specified.
    `..' is processed by removing the immediately previous pathname component
    back to a slash or the beginning of DIR.
```

Рисунок 10: Справка по команде cd


```
PWD(1)                                User Commands                                PWD(1)

NAME
    pwd - print name of current/working directory

SYNOPSIS
    pwd [OPTION]...

DESCRIPTION
    Print the full filename of the current working directory.

    -L, --logical
        use PWD from environment, even if it contains symlinks

    -P, --physical
        resolve all symlinks

    --help display this help and exit

    --version
        output version information and exit

    If no option is specified, -P is assumed.

    Your shell may have its own version of pwd, which usually supersedes the version described here.
    Please refer to your shell's documentation for details about the options it supports.

AUTHOR
    Written by Jim Meyering.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Manual page pwd(1) line 1 (press h for help or q to quit)
```

Рисунок 11: Справка по команде pwd

```
MKDIR(1)                                User Commands                                MKDIR(1)

NAME
  mkdir - make directories

SYNOPSIS
  mkdir [OPTION]... DIRECTORY...

DESCRIPTION
  Create the DIRECTORY(ies), if they do not already exist.

  Mandatory arguments to long options are mandatory for short options too.

  -m, --mode=MODE
      set file mode (as in chmod), not a=rwx - umask

  -p, --parents
      no error if existing, make parent directories as needed, with their file modes unaffected by any
      -m option

  -v, --verbose
      print a message for each created directory

  -Z      set SELinux security context of each created directory to the default type

  --context[=CTX]
      like -Z, or if CTX is specified then set the SELinux or SMACK security context to CTX

  --help display this help and exit

  --version
      output version information and exit

Manual page mkdir(1) line 1 (press h for help or q to quit)
```

Рисунок 12: Справка по команде mkdir

```
RMDIR(1)                                User Commands                                RMDIR(1)

NAME
    rmdir - remove empty directories

SYNOPSIS
    rmdir [OPTION]... DIRECTORY...

DESCRIPTION
    Remove the DIRECTORY(ies), if they are empty.

    --ignore-fail-on-non-empty
        ignore each failure to remove a non-empty directory

    -p, --parents
        remove DIRECTORY and its ancestors; e.g., 'rmdir -p a/b' is similar to 'rmdir a/b a'

    -v, --verbose
        output a diagnostic for every directory processed

    --help display this help and exit

    --version
        output version information and exit

AUTHOR
    Written by David MacKenzie.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>

Manual page rmdir(1) line 1 (press h for help or q to quit)
```

Рисунок 13: Справка по команде rmdir

```
RM(1)                                User Commands                                RM(1)

NAME
  rm - remove files or directories

SYNOPSIS
  rm [OPTION]... [FILE]...

DESCRIPTION
  This manual page documents the GNU version of rm.  rm removes each specified file.  By default, it does not remove directories.

  If the -i or --interactive=once option is given, and there are more than three files or the -r, -R, or --recursive are given, then rm prompts the user for whether to proceed with the entire operation.  If the response is not affirmative, the entire command is aborted.

  Otherwise, if a file is unwritable, standard input is a terminal, and the -f or --force option is not given, or the -i or --interactive=always option is given, rm prompts the user for whether to remove the file.  If the response is not affirmative, the file is skipped.

OPTIONS
  Remove (unlink) the FILE(s).

  -f, --force
        ignore nonexistent files and arguments, never prompt

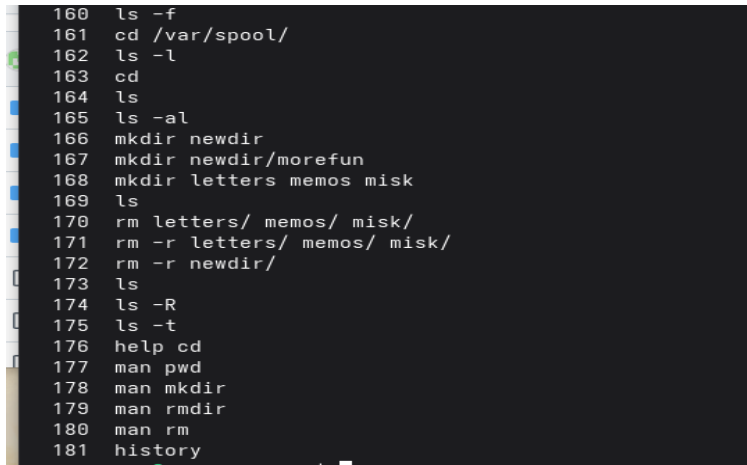
  -i
        prompt before every removal

  -I
        prompt once before removing more than three files, or when removing recursively; less intrusive than -i, while still giving protection against most mistakes

  --interactive[=WHEN]

Manual page rm(1) line 1 (press h for help or q to quit)
```

Рисунок 14: Справка по команде rm



```
160 ls -f
161 cd /var/spool/
162 ls -l
163 cd
164 ls
165 ls -al
166 mkdir newdir
167 mkdir newdir/morefun
168 mkdir letters memos misk
169 ls
170 rm letters/ memos/ misk/
171 rm -r letters/ memos/ misk/
172 rm -r newdir/
173 ls
174 ls -R
175 ls -t
176 help cd
177 man pwd
178 man mkdir
179 man rmdir
180 man rm
181 history
```

Рисунок 15: Команда history

3. Выводы по проделанной работе

Мы приобрели практические навыки взаимодействия пользователя с системой посредством командной строки.